

Aspects of *-ki* Nominalization in Korean*

Michael Barrie** · Heeryun Chung***
(Sogang University)

Michael Barrie · Heeryun Chung. 2019. Aspects of *-ki* Nominalization in Korean. *Language and Linguistics* 83, 1-44. This paper analyzes *-ki nominalizations* in Korean. We identify two types of the *-ki nominalizations*, NOM-*ki* and GEN-*ki*, with distinct morphosyntactic properties. We propose an analysis along the lines of Kratzer (1996) in which the nominalizer *-ki* selects verbal projections of different sizes (TP and vP) to account for the observed patterns. We argue against previous accounts of *-ki* nominalizations, showing either that important generalizations were missed or that there were flaws in the previous analyses. We propose a novel generalization for the two types of *-ki* nominalizations and propose an appropriate analysis. Some outstanding issues and implications are briefly discussed.

Keywords: nominalization, Korean, Case marking, noun incorporation, tense in nominals

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** The first author: Michael Barrie

*** The corresponding author: Heeryun Chung

1. Introduction

We analyze clausal *-ki* nominalizations in Korean accounting for interesting correlations between tense and Case. In a nutshell, there are two kinds of *-ki* nominalizations in Korean akin to the three kinds of *-ing* nominalizations in English, as discussed by Abney (1987) and Kratzer (1996). One type, which we call NOM-*ki*, adapting the same kind of nomenclature used for *-ing* nominalizations in English, has nominative subjects and accusative objects. The other type, GEN-*ki*, has genitive subjects and Caseless objects. The properties of these two types of nominalizations are remarkably similar to those found in English, and we propose a similar analysis. While English has *of*-insertion to account for the lack of accusative Case in certain *of-ing* constructions, we propose that Korean makes use of a kind of noun incorporation, in the sense of Barrie & Mathieu (2016) and Lee (2018).

The analysis in a nutshell runs as follows. The NOM-*ki* construction has nominative subjects, accusative objects, tense marking, and adverbial modification only. We argue that such constructions involve a TP being nominalized by *-ki*. The GEN-*ki* construction has genitive subjects, caseless objects, no tense marking, and adverbial or adnominal modification. We argue that such constructions are the result of a bare vP being nominalized by *-ki*. We discuss some problematic issues such as the presence of an agentive external argument in the GEN-*ki* construction despite the lack of VoiceP, the head that is responsible for introducing the external argument and for checking accusative Case. We tentatively suggest that Moulton's (2004) proposal for a similar phenomenon in English might account for the Korean facts, but we leave the precise implementation to future work.

The remainder of this paper is structured as follows. Section 2 provides the background necessary for this study, including the properties of *-ki* nominalization and the theoretical background on Kratzer's VoiceP, her analysis on *-ing* nominalizations in English, and Marantz's proposed analysis of acategorial roots and categorizers. Section 3 presents in detail the properties of the two types of *-ki* nominalizations. Section 4 presents previous analyses of *-ki* nominalization in Korean, pointing out shortcomings in these analyses. Section 5 presents our analysis. Section 6 is a brief conclusion.

2. Background

This section provides the background on Korean nominalizations for the forthcoming proposal. First, we present *-ki* nominalizations, in which two different Case possibilities are possible for the subject and object. We introduce these here briefly and then explicate them fully in section 3. Second, we provide the theoretical background necessary for our analysis. In particular, we discuss Kratzer's (1996) analysis of nominalization and Marantz's (2001) analysis of categorization.

2.1. Background on Korean *-ki* Nominalization

Both *-ki* nominalizations in (1) contain a subject, direct object, and nominalized verb. The *-ki* nominalization in (1a) appears with nominative Case, while the one in (1b) appears with genitive Case. Focusing on this fact, in this paper, we classify *-ki* nominalizations with respect to the type of Case they appear with on the subject.

For ease of exposition, we use the term *NOM-ki* when the subject appears with nominative Case and *GEN-ki* for those with a genitive Case, akin to the terminology used for *-ing* nominalizations in English.¹⁾

- (1) a. Yeungi-nun [Minswu-ka chayk-ul ilk]-ki-lul
 Yeungi-TOP Minswu-NOM book-ACC read-NML-ACC
 wenha-n-ta.
 want-PRES-DEC
 'Yeungi wants Minswu to read books.'
- b. [Minswu-uy chayk ilk]-ki-ka sicak-toy-ess-ta.
 Minswu-GEN book read-NML-NOM start-PASS-PAST-DEC
 'Minswu's reading a book has started.'

It will be shown that these two different types of *-ki* nominalization in (1) possess the following syntactic asymmetries: the Case on the direct object, availability of tense, and type of modification (adverbial or adnominal). Crucially, these asymmetries help elucidate the syntactic structures of the two types of *-ki* nominalizations in Korean.

1) A reviewer notes that there are several examples in which *GEN-ki* is ungrammatical. They note, furthermore, that such examples in corpora are hard to come by if at all. While we agree that *GEN-ki* constructions are rare, they are not altogether ruled out. The following example is completely natural.

i) (Chelswu-uy yehayng ka)-ki kyeyhoyk
 Chelswu-gen vacation go-nml plan
 'Chelswu's plan to go on a vacation'

One topic we leave for future research is why some *GEN-ki* constructions are ungrammatical (as the reviewer notes) while others are perfectly grammatical. We consider here only the morphosyntactic properties of *-ki* nominalizations and leave the important question of the restrictions on the *GEN-ki* for future research.

2.2. Theoretical Background

2.2.1. Kratzer's (1996) VoiceP

Kratzer (1996) argues that the external argument is introduced by a functional head, Voice, denoting a thematic relation between the external argument and the event described by the verb. The Voice head is located above VP with the possibility of intervening inflectional heads. The object of the verb, then, moves further up to receive accusative Case (Kratzer 1996:126). While we adopt Kratzer's influential proposal that the external argument is "severed" from the verb's argument structure, we further adopt the more recent proposal that the Voice head proposed by Kratzer (and nearly simultaneously by Chomsky (1995)) is split into two projections as described in the next section.

Kratzer suggests that gerunds are a window into the hierarchy of extended projection of verbs. Extending and more articulating Abney's (1987) argument that the English nominalizing affix *-ing* can attach to Vs, VPs, or IPs, she proposes that gerunds are formed by nominalizing different segments of a verb's extended projection, and furthermore that such nominalizations can diagnose the presence or absence of Voice in the nominalization.

Three kinds of *-ing* nominalizations in English, *of-ing*, *POSS-ing*, and *ACC-ing*, are described as follows. In *of-ing* gerunds (2a), the absence of accusative Case corresponds to the absence of the verb's external argument. Thus, the nominalized constituent does not contain Voice, but instead, is at most a VP (under Kratzer's original proposal, which we revise slightly below). As for the *POSS-ing* (2b) and *ACC-ing* (2c) gerunds, the presence of accusative Case is accompanied by the presence of the external argument, and this

leads us to the conclusion that the nominalized constituent includes Voice.

- (2) a. Maria's reading of Pride and Prejudice
- b. Maria's reading Pride and Prejudice
- c. Maria enjoyed reading Pride and Prejudice.

(adapted from Kratzer 1996)

Kratzer (1996) connects her Voice head analysis to Johnson's (1991) functional head μ , which assigns accusative Case to the direct object. According to Johnson, the noun phrase moves to Spec of VP.²⁾ Therefore, they precede adverbs and other complements of verbs. Consider the following examples below.

- (3) a. Mittie fed the dog quickly.
- b. *Mittie fed quickly the dog.
- c. Mikey visited his parents quietly.
- d. *Mikey visited quietly his parents.
- e. Gary told Sam to leave.
- f. *Gary told to leave Sam.

(adapted from Johnson 1991)

In (3), the sentence is ungrammatical if the object noun phrase does not move to the Spec of VP, while noun phrases that undergo this movement are grammatical. We do not recapitulate Johnson's original arguments, but rather refer the reader to Johnson (1991) to see how the linear order in (3) relates to the structure. The movement of object NP will support our account for modification asymmetry in section 5.3.

2) We use the term *noun phrase* to express ambivalence as to whether the nominal is a bare NP, a DP or a KP (or any other possible nominal structure).

2.2.2. Marantz' s (2001) Categorizers

According to Marantz (2001), words are built syntactically, starting with acategorial roots. Specifically, a categorizing head, whose function is to define the lexical category of the word (N, V, Adj) is merged with the root before merging other functional heads. The categorizing heads are identified as *v*, *n*, and *a*. To give an English example, affixes such as *-tion*, *-ity*, *-age* are *n* heads and are merged with a base or with a root to form a noun (Marantz, 2001). Following Newell (2008), Harley (2013), and Legate (2014) we distinguish between Marantz's categorizing *v* head and Kratzer's Voice head, the latter of which is responsible for assigning accusative Case and for introducing the external argument.

Following Marantz's idea, we argue that both types of *-ki* nominalizations in (1) are contained in an *nP* projection. Specifically, the nominalizer *-ki* is one of the category-determining morphemes instantiated by *n*. In the next section, we discuss properties of *-ki* nominalization and describe three asymmetries correlated with the difference in Case on the subject.

3. Properties of Korean Nominalization

In addition to *-ki*, verbal categories can also be nominalized by *-(u)m*.³⁾ Although we analyze only *-ki* nominalizations here, several previous studies compared both of these nominalizers, so we address both of them in this section, leaving a detailed study of

3) Clausal nominalizations are also formed with *-kes*, and lexical nominalization are formed with a variety of suffixes such as *-kay* and *-i* (Yeon & Brown, 2011). The syntactic properties of these other nominalizations are significantly different from *-ki* nominalizations, so we set these aside here.

-(*u*)*m* to future work. The choice between *-ki* and *-(u)m* is based on the subcategorizational characteristics of the matrix verb (Yoon 1989; Song 1990; Choi 2015).⁴⁾ Consider the following examples (Yoon 1989).

- (4) a. [John-i hakkyo-ey ka]-ki-ka swip-ta.
 John-NOM school-to go-NML-NOM easy-DEC
 'It is easy for John to go to school.'
- b. *[John-i hakkyo-ey ka]-m-i swip-ta.
 John-NOM school-to go-NML-NOM easy-DEC
 'It is easy for John to go to school.'
- c. John-un [Mary-ka hakkyo-ey ka]-m-ul po-ass-ta.
 John-TOP Mary-NOMschool-to go-NML-ACC see-PST-DEC
 'John saw Mary go to school.'
- d. *John-un [Mary-ka hakkyo-ey ka]-ki-lul po-ass-ta.
 John-TOP Mary-NOMschool-to go-NML-ACC see-PST-DEC
 'John saw Mary go to school.'
- (adapted from Choi 2015, citing Yoon 1989)

When the matrix verb is *swip-ta* (easy), the verb *ka* (go) takes the affix *-ki* as its nominalizer in (4a), while *-(u)m* is rejected (4b). On the other hand, when the matrix verb is *po-ta* (see), the *-(u)m* nominalizer is added (4c) rather than the *-ki* nominalizer (4d).

3.1. Asymmetry 1: The Case on the Direct Object

In *-ki* nominalizations with NOM-*ki*, as in (5a), accusative Case assignment on the direct object is obligatory. However, when a *-ki* nominalized clause appears with a GEN-*ki*, as in (5b), the direct object does not appear with accusative case. It remains without any

4) In particular the occurrence of *-ki* and *-(u)m* is restricted by the semantic characteristics of matrix predicates, which has been described in terms of future-orientedness, realis mood, existence, decisiveness, and volition, depending on the author (Lim 1979; Song 1990).

case marker. This is the first asymmetry with *-ki* nominalization.⁵⁾

- (5) a. Mina-nun [Yunu-ka pomwul*(-ul) chac]-ki-lul
 Mina-TOP Yunu-NOM treasure-ACC search-NML-ACC
 pala-n-ta.
 hope-PRES-DEC
 'Mina wants Yunu to search for treasure.'
 'Mina wishes that Yunu finds treasure.'
- b. Mina-nun [Yunu-uy pomwul(*-ul) chac]-ki-lul
 Mina-TOP Yunu-GEN treasure search-NML-ACC
 towa-ss-ta.
 help-PAST-DEC
 'John helped Mary's treasure hunt.'

Interestingly, the accusative case on the direct object in (6) is optional when the light verb *ha* takes the nominalized verb as its complement despite the nominative case on the subject (*compare* with (5a)). Based on the data in (5) and (6), we assume that the core ingredients for the first asymmetry are caused by the type of nominalized verb as well as nominative and genitive cases on the external argument.

- (6) na-nun[Mina-ka [pomwul-(*ul) chac]-ki-(lul) ha]-ki-lul
 I-TOP Mina-NOM treasure search-NML-ACC v-NML-ACC
 pala-n-ta.
 hope-PRES-DEC
 'I want Mina to search for treasure.'

5) A reviewer notes that examples such as (5a) are not fully ungrammatical when accusative case is absent. We have asked several native speakers of Korean and most have agreed that when accusative case is absent the sentence is ungrammatical or quite degraded. Only one speaker has said that the sentence is only mildly degraded. In future work we will conduct Likert Scale test to investigate this kind of construction in more detail.

This asymmetry is summarized in the following schema.

(7) [Asymmetry 1] Case Marking on the Direct Object

- a. [Subject_{NOM} Object_{ACC} V]-*ki*
- b. *[Subject_{NOM} Object-Ø V]-*ki*
- c. *[Subject_{GEN} Object_{ACC} V]-*ki*
- d. [Subject_{GEN} Object-Ø V]-*ki*

3.2. Asymmetry 2: Tense

The second asymmetry in *-ki* nominalizations lies in tense. As shown in (8a), the past tense morpheme *-ass* appears freely in *-ki* nominalizations with a nominative subject. On the contrary, (8b) shows that past tense is unavailable in *-ki* nominalizations with a genitive subject.

- (8) a. Mina-nun [Yunu-ka pomwul-ul chac-(ass)]-ki-lul
 Mina-TOP Yunu-NOM treasure-ACC search-PAST-NML-ACC
 pala-n-ta.
 hope-PRES-DEC
 'Mina wishes that Yunu finds/had found the treasure.'
- b. Mina-nun [Yunu-uy pomwul chac-(*ass)]-ki-lul
 Mina-TOP Yunu-GEN treasure search-PAST-NML-ACC
 top-nun-ta.
 help-PRES-DEC
 'Mina is helping with Yunu's finding of the treasure.'

A reviewer noted that the examples in (8) differ not only in tense in the nominalized clause but also in the matrix verb. Because of this, we include a better minimal pair from CITATION.

- (9) a. Chelswu-nun [Yenghuy-ka sihem-ul cal
 Chelswu-TOP Yenghuy-NOM exam-ACC well
 po-(ass)]-ki-lul pala-ss-ta.
 take-PAST-NML-ACC hope-PAST-DEC
 'Chelswu hoped that Yenghuy took the exam well.' (sic)
- b. Chelswu-nun [Yenghuy-uy sihem-ul cal
 Chelswu-TOP Yenghuy-GEN exam-ACC well
 po-(*ass)]-ki-lul pala-ss-ta.
 take-PAST-NML-ACC hope-PAST-DEC
 'Chelswu hoped that Yenghuy took the exam well.' (sic)
- (Kang 1988:87(5) and (7))

The second asymmetry is summarized as follows. Tense morphology can appear in *-ki* nominalizations with a nominative subject, but not with a genitive subject.

- (10) [Asymmetry 2] Tense
- a. [Subject_{NOM} Object_{ACC} V_(past)]-*ki*
- b. [Subject_{GEN} Object_{ACC} V_(*past)]-*ki*

3.3. Asymmetry 3: Modification

The third asymmetry is found in the possibility of adnominal or adverbial modifiers appearing in *-ki* nominalizations. Consider first the following examples in (11), which instantiate *-ki* nominalization with NOM-*ki*. Adverbial modification *ppalu-key* (fast-ADV) is allowed in the *-ki* nominalization with NOM-*ki*, as in (11a-b), while adnominal modification *ppalu-n* (fast-ADN) is not allowed (11c-d), regardless of word order.⁶⁾

6) The examples in (11) to (12) are not presented as full sentences but as nominalized phrases to show the acceptability of modifiers inside nominalizations. We present the examples of *-ki* nominalization with GEN-*ki* (12), without accusative case marker *-(l)ul* on the internal argument *sakwa* (apple) as we already have discussed in (5). Thirty-five participants whose

- (11) a. Mina-ka ppalu-key sakwa-lul mek-ki
 Mina-NOM quick-ADV apple-ACC eat-NML
 ‘Mina quickly eating an apple’
 b. Mina-ka sakwa-lul ppalu-key mek-ki
 Mina-NOM apple-ACC quick-ADV eat-NML
 ‘Mina quickly eating an apple’
 c. *Mina-ka ppalu-n sakwa-lul mek-ki
 Mina-NOM quick-ADN apple-ACC eat-NML
 ‘Mina’s quick eating an apple’
 d. *Mina-ka sakwa-lul ppalu-n mek-ki
 Mina-NOM apple-ACC quick-ADN eat-NML
 ‘Mina’s quick eating an apple’

In contrast to *-ki* nominalization with NOM-*ki* (11c), *-ki* nominalization with GEN-*ki* allows adnominal modification (12c) as well as adverbial. Note, however, that there is a word order restriction with adnominal modification (compare (11c-d) and (12c-d)). The adnominal modifier cannot appear lower than the object, (12d). This is the last asymmetry related to different Case marking on the subject.

- (12) a. Yunu-uy ppalu-key sakwa mek-ki
 Yunu-GEN quick-ADV apple eat-NML
 ‘Yunu’s quickly eating an apple’
 b. Yunu-uy sakwa ppalu-key mek-ki
 Yunu-GEN apple quick-ADV eat-NML
 ‘Yunu’s quickly eating an apple’
 c. Yunu-uy ppalu-n sakwa mek-ki
 Yunu-GEN quick-ADN apple eat-NML
 ‘Yunu’s quick eating an apple’

native language is Korean conducted a grammaticality judgment task for the examples in (11) and (12). They judged the phrases in (11c-d) and (12d) to be completely ungrammatical with a score of 1 out of 7 (under Likert Scale test), while the rest of (11) and (12) were judged with a score of 5 (12a-c) or 7 (11a-b).

- d. *Yunu-uy sakwa ppalu-n mek-ki
 Yunu-GEN apple quick-ADN eat-NML
 ‘Yunu’s quick eating an apple’

The following two schemata summarize these generalizations.

(13) [Asymmetry 3] Modification Pattern with adnominals

- a. Subject_{NOM} (*adn) Object_{ACC} (*adn) V-*ki*
 b. Subject_{GEN} (adn) Object_Ø (*adn) V-*ki*

(14) [Asymmetry 3] Modification Pattern with adverbials

- a. Subject_{NOM} (adv) Object_{ACC} (adv) V-*ki*
 b. Subject_{GEN} (adv) Object_Ø (adv) V-*ki*

3.4. Interim Summary

To summarize, we have discussed three asymmetries with *-ki* nominalizations in Korean. The first asymmetry is the presence or absence of Case on the direct object, which has, to date, largely been ignored in the literature.⁷⁾ The second asymmetry concerns the possibility of tense marking. The third asymmetry lies in modification patterns (adnominal versus adverbial). The clearest point is that all the above mentioned asymmetries are correlated with Case on the external argument (NOM vs. GEN). To the best of our knowledge, these three asymmetries have not been addressed under the umbrella of a single analysis before.

7) There has been much attention paid to Case in Korean nominalizations in that they possess argument structures, especially, genitive case drop (An, 2009) and case assignment system in comparing between *-ki* and *-(u)m* nominalization (Chung, 2014). As far we know, however, asymmetries on the case assignment when nominalizations appear as embedded objects of matrix clauses has not been studied yet.

〈Table 1〉 Summary of Asymmetries

Asymmetry	Nominative Subject	Genitive Subject
Case on Direct Object	Accusative	none
Tense	present or past	present only
Modification	ADV only	adjacent to V: ADV or ADN elsewhere: ADV only

One may ask whether these asymmetries also found if intransitive or causative verbs are nominalized. Preliminary results in Chung (forthcoming) suggest that the picture is complicated by verb type and adicity (transitive, unergative, unaccusative, passive, causative, inchoative), but we do not discuss the details here. In short, the main idea of Chung is that there is an interaction between verb type and the ability to undergo nominalization (see also footnote 8).

4. Previous Analyses

In this section, we present previous studies comparing *-ki* and *-(u)m* nominalizations (Kim 2000, 2005; Jeong 2004; Yoon 2018) and discuss the insights they have made. Especially, we point out their problems relevant to *-ki* nominalizations. We also lay out the structural analyses proposed (Han 1987; Yoon 1989b; Chung 2014) and the problems that have been pointed out (Choi, 2015).

4.1. Kim (2000, 2005): Difference between *-ki* and *-(u)m*

Kim (2000) discusses three syntactic differences between *-ki* and *-(u)m* affixation: case, modifiers (adnominal or adverbial), and

tense. First, Kim (2000) observes that *-(u)m* can take a clause with a genitive subject but *-ki* cannot, as presented in (15). In (15d), the sentence is ungrammatical when the embedded subject appears with a genitive case marker.

- (15) a. na-nun [kunye-ka tolao]-m-ul pala-n-ta.
 I-TOP she-NOM return-NML-ACC hope-PRES-DEC
 'I hope that she returns.'
- b. na-nun [kunye-uy tolao]-m-ul pala-n-ta.
 I-TOP she-GEN return-NML-ACC hope-PRES-DEC
 'I hope for her return.'
- c. na-nun [kunye-ka tolao]-ki-lul pala-n-ta.
 I-TOP her-NOM return-NML-ACC hope-PRES-DEC
 'I hope that she returns.'
- d. *na-nun [kunye-uy tolao]-ki-lul pala-n-ta.
 I-TOP she-GEN return-NML-ACC hope-PRES-DEC
 'I hope for her return.'
- (adapted from Kim 2000:60 (15) and (16))

Note that in Kim's data in (15), the nominalized verb *tolao-ta* (return) is an intransitive verb. When the nominalized verb is transitive *chac-ta* (search), however, a subject marked with a genitive case is allowed, as in (16b). Thus, Kim's claim that *-ki* cannot take a clause with a genitive marked subject does not hold.⁸⁾ The subject of a transitive verb nominalized with *-ki* can appear with genitive case.

- (16) a. na-nun [Minswu-ka pomwul-ul chac]-ki-lul pala-n-ta.
 I-TOP Minswu-NOM treasure-ACC seek-NML-ACC hope-PRES-DEC
 'I hope for Minswu to seek out the treasure.'

8) A cursory investigation of the data suggests that only transitive verbs can take a genitive subject in *-ki* nominalized clauses. Chung (forthcoming) argues that unergative and unaccusative verbs cannot be nominalized with a genitive subject.

- b. na-nun [Minswu-uy pomwul chac]-ki-lul
 I-TOP Minswu-GEN treasure seek-NML-ACC
 top-nun-ta.
 help-PRES-DEC
 'I help with Minswu's seeking out the treasure.'

We thus revise Kim's (2000) claim and conclude that *-ki* nominalized verbs can appear with either a nominative or a genitive subject; however, it appears that only transitive verbs nominalized with *-ki* can take genitive a subject. As mentioned, the interaction between adicity and nominalization is discussed in Chung (forthcoming).

Second, Kim (2000) considers modifiers in the nominalized clauses. Based on the examples in (17), she generalizes that *-(u)m* nominalization can be modified by only adverbs when its subject is marked with a nominative case (17a), while it can be modified by either adnominals or adverbs when its subject is marked with a genitive case (17b).

- (17) a. [Chulswu-ka *ilu-n/ilu-key ttena]-m-i
 Chulswu-NOM quick-ADN/quick-ADV leave-NML-NOM
 sepsep-ha-ta.
 disappointed-v-DEC
 'Chulswu's leaving quickly is disappointing.'
- b. [Chulswu-uy ilu-n/ilu-key ttena]-m-i
 Chulswu-GEN quick-ADN/quick-ADV leave-NML-NOM
 sepsep-ha-ta.
 disappointed-v-DEC
 'Chulswu's leaving quickly is disappointing.'
- (adapted from Kim 2000:61)

However, Kim's arguments about adverbs in *-(u)m* nominalization with GEN-*ki* are in opposition to Yoon's (2018) argument. According to Yoon, adverbial modification appearing with GEN-*ki* is

- b. ku-nun [na-eykey *ppallu-n/ppal-li
 he-TOP I-DAT quick-ADN/quick-ADV
 chwulsey-ha]-ki-lul pala-n-ta.
 succeed-v-NML-ACC hope-PRES-DEC
 'He hopes that I quickly succeed.'

(adapted from Kim 2000:62)

Kim (2000) tries to establish syntactic differences between *-ki* and *-(u)m* nominalizations with the data set above, however, there are problems as follows. First, she does not consider nominalization with GEN-*ki* to compare the modification system of *-ki* and *-(u)m* nominalizations, as she erroneously assumes that *-ki* nominalized clauses cannot appear with GEN-*ki*. Instead, she considers the embedded dative subject, as in (20b). Therefore, we cannot see whether there is a difference between *-ki* and *-(u)m* nominalizations with respect to the pattern of modification. Second, even if we compare (17a) and (20a) excluding the examples with dative case, we cannot find any gaps between them since both allow adverbs but not adnominals. Nevertheless, Kim (2000) mistakenly concludes that *-(u)m* nominalization can take either adnominals or adverbs when its subject appears with a genitive case (*contra* Yoon 2018) while *-ki* nominalization can take only adverbs (20a).

When a broader range of data is considered, however, we see that genitive subjects are possible. Consider Yoon's (2018) data with respect to modification in *-ki* nominalization with GEN-*ki* in (21).

- (21) a. haksayngtul-uy *ppalu-key/ppalu-n talli-ki
 students-GEN quick-ADV/quick-ADN run-NML
 '(the) students' quick running'
 b. hachengepca-uy *cayppalu-key/cayppalu-n mek-ko-thwi-ki
 subcontractor-GEN quick-ADV/quick-ADN eat-CMP-take.off-NML
 '(the) subcontractor's quick dining and dashing'

(adapted from Yoon 2018:(29) and (35))

As we have seen so far, there are two different observations (Kim's and Yoon's) for adverbial modifications in *-(u)m* nominalizations. Recall that our generalizations for (11) to (12) obtained the experiment on thirty-five Korean native speakers: i) in the nominalization with *NOM-ki*, only adverbial modification is grammatical and ii) in the nominalization with *GEN-ki*, both adnominal and adverbial modifications are allowed, except the intervening adnominal between the direct object and the nominalized verb. In this paper, we argue that adverbial modification is also allowed (*contra* Yoon 2018) inside the nominalization with a genitive Case, based on our observations as we presented in (11) and (12). The vital point is that the modification asymmetry clearly exists depending on the case marking of external arguments in nominalizations regardless of nominalizers.

Kim (2000, 2005) also compares *-ki* and *-(u)m* with respect to tense possibilities. She states that verbs nominalized with *-(u)m* can appear with past tense (22a) while verbs nominalized with *-ki* cannot (22b).

- (22) a. [ku-eykey mwuncey-ka iss-(ess)-um]-ul
 he-DAT problem-NOM be-PAST-NML-ACC
 cicek-ha-yess-ta.
 point out-v-PAST-DEC
 'Someone pointed out that he has/had a problem.'
- b. [kutul-un selo manna-(*ss)]-ki-lul kkely-ess-ta.
 they-TOP each other see-PAST-NML-ACC unwilling-PAST-DEC
 'They were unwilling to see each other.'
- (adapted from Kim 2005:25)

We believe the contrast in (22) is spurious, however. Note that Jeong (2004) observes that *-ki* and *-(u)m* nominalizations are in

complementary distribution in terms of the types of matrix predicates they combine with. For instance, the *-ki* nominalizer can take past tense when it appears with [+expectancy] types of verbs such as *pala-ta* (hope), while *-(u)m* cannot (23a). On the contrary, the *-ki* nominalizer cannot take past tense when the matrix verb is *cwucangha-ta* (claim) types of verbs which involve [+perfective], while *-(u)m* can, (23b). This leads us to assume that the difference in the availability of past tense between *-ki* and *-(u)m* is related to the semantic properties of the verb.

- (23) a. na-nun ai-ka cal cala-(ss)-*ki*/**um*-ul pala-n-ta.
 I-TOP child-NOM well grow-PAST-NML-ACC hope-PRES-DEC
 'I hope that a child grows/grew up well.'
- b. Yeungi-nun ku-ka coy-ka eps-(ess)-*um*/**ki*-lul
 Yeungi-TOP he-NOM guilt-NOM not.have-PAST-NML-ACC
 cwucangha-n-ta.
 claim-PAST-DEC
 'Yeungi claimed that he is/was free from guilt.'
- (adapted from Jeong 2004)

Thus, we argue that tense asymmetry arises from not the difference between *-ki* and *-(u)m* but the case marking of external argument (NOM vs. GEN) as we discussed in section 2. The nominalization with NOM-*ki* takes past tense in both *-ki* and *-(u)m* while GEN-*ki* does not.

In sum, we have seen that Kim's (2000) generalizations on the properties of *-ki* and *-(u)m* nominalizations and pointed out several errors on them. Based on these, we conclude as follows. First, the three syntactic aspects (case, modifiers, and tense) suggested by Kim (2000) have very little to contribute to the difference between nominalizer *-ki* and *-(u)m*. Instead, it is clear that the occurrence of the nominalizer (at least *-ki* and *-(u)m*) in Korean depends on

the matrix predicate (Lim 1979; Song 1990; Jeong 2004). Second, there are two different structures determined by case marking on the external argument.

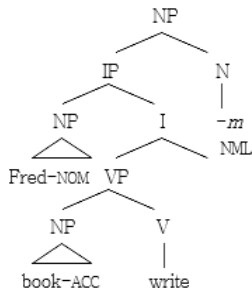
4.2. Previous Structural Analyses

For the forthcoming proposal, we present some previous treatments of Korean nominalization structures and various problems that have been pointed out. As we have seen, the subject inside the nominalization in Korean can take either genitive or nominative Case. In the following subsections, we concentrate on structures of Korean nominalization with nominative Case (Han 1987; Yoon 1989) and genitive Case (Chung 2014), respectively.

4.2.1. Nominalization with the Nominative Case

Han (1987) argues that the nominalizers *-ki* and *-(u)m* have the same properties syntactically and proposes that they are N heads that take IPs as their complements. Han's structure for the sentential nominalization is presented in below:

- (24) Fred-ka chayk-ul ssu-m
 Fred-NOM book-ACC write-NML
 'That Fred wrote a book' (sic)



(Han 1987, cited in Choi 2015)

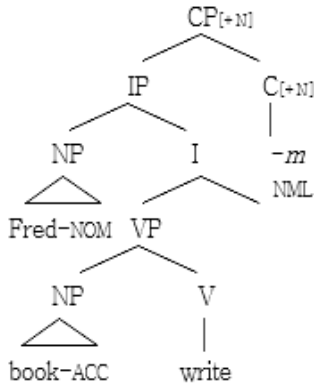
The structure in (24) shows the internal structure of Korean sentential nominalization. However, Choi (2015) points out that this structure is exactly the same as that of a complex noun phrase (CNP), and thus should be subject to the CNP constraint (Ross, 1967). Choi shows that this is not the case, however. Consider the examples in (25).

- (25) a. **pap*_i-un Kim-i [_{NP} [_{IP} Tom-i t_i mek-ess-ta-nun]
 meal-TOP Kim-NOM Tom-NOM eat-PAST-DEC-REL
 sasil]-ul palkhi-ess-ta.
 fact-ACC reveal-PAST-DEC
 'As for rice_i, Kim revealed the fact that Tom ate it_i.'
- b. *Yenge*_j-nun Kim-i [_{e_i} t_j kongpu-ha-ki-lul] coha-ha-n-ta.
 English-TOP Kim-NOM study-v-NML-ACC like-v-PRES-DEC
 'As for English, Kim likes studying it.'
- (Choi 2015:75 (7))

The complex noun phrase in (25a) is ungrammatical since *pap* (meal) is extracted out of the CNP. In (25b), on the other hand, *yenge* (English) is extracted out of a nominalized phrase and the sentence is completely acceptable.

Yoon (1989) argues that nominalization affixes *-ki* and *-(u)m* occupy CP head position as shown in (26). The nominalization affix is marked with the [+N] feature in the C head position. The CP with a [+N] feature shows the external nominal property. Yoon argues that this approach avoids the CNPC violation, unlike Han's approach.

- (26) Fred-ka chayk-ul ssu-m
 Fred-NOM book-ACC write-NML
 'That Fred wrote a book.' (sic)



(Yoon 1989, cited in Choi 2015)

However, Yoon's (1989) approach also has a problem as pointed out by Choi (2015). Consider the examples in (27). The matrix verb *keylsimhata* (determine) in (27a) generally takes a nominalized phrase as its complement omitting the subject of the nominalized verb. If there is a subject inside the nominalized phrase as in (27b), the sentence is unacceptable. Yoon's approach, however, cannot explain the obligatory absence of the subject in the nominalization phrase as a Spec of IP position should always be available.

- (27) a. John-un [kongpwu-lul ha]-ki-lul kyelsim-ha-yess-ta.
 John-TOP study-ACC do-NML-ACC determine-v-PAST-DEC
 'John is determined to study.' (sic)
 b. *John-un [Kim-i kongpwu-lul ha]-ki-lul
 John-TOP Kim-NOM study-ACC do-NML-ACC
 kyelsim-ha-yess-ta.
 determine-v-PAST-DEC
 'John determined Kim to study.' (sic)

(Choi 2015:77 (10))

Yoon's (1989) approach poses other problems, which Choi (2015) points out. First, it is unclear what the difference is between $CP_{[+N]}$ and NP. If there is no clear difference between them, Yoon's approach also confronts the same problem that Han (1987) encounters, namely a CNPC violation. Second, If the $CP_{[+N]}$ differs from the NP, the question is what the $[+N]$ feature is.

We have seen previous syntactic approaches with Korean sentential nominalization and problems with them. As a result, we reject Han's (1987) 'NP-IP' and Yoon's (1989) ' $CP_{[+N]}$ -IP' approaches in that both are not able to explain above mentioned CNPC violation. Also as mentioned the properties of feature $[+N]$ is not clear. We now turn to Chung's (2014) approach to the structure of nominalization marked with the genitive case *-uy* instead of a nominative case.

4.2.2. Nominalization with the Genitive Case: *vP* (Chung 2014)

Chung (2014) mentions that there are two types of *-(u)m* nominalizations in Korean, phrasal nominalization and lexical nominalization. Consider the following examples that Chung provides.

- (28) a. $[_{IP}$ John-i $[_{vP}$ tosi-lul phakoy-ha-yess]]-um
 John-NOM city-ACC destruction-v-PAST-NML
 'That John completely destroyed the city'
 b. John-uy $[_v$ cwuk]-um
 John-GEN die-NML
 'John's death'

(Chung 2014:18)

In (28a), the *-(u)m* nominalization appears with the accusative case marker *-lul* on the internal argument *tosi* (city). In (28b),

-(u)m nominalization does not show any verbal case marking. The former targets the phrasal structures (*vP* and the *IP*), while the latter targets the lexical head *V*, following Yoon (1996).

Assuming that the accusative Case is assigned by the light verb, *ha* in Korean has been argued to be *v* (Ahn 1991; Chung 2007; Park, 2007). When the verbal noun *phakoy* (destruction) which always requires *ha* does not have it, as in (29a), the accusative Case *-lul* must be dropped. If the verbal noun possesses *ha*, accusative Case assignment is obligatory (29b). This shows that there is an apparent correlation between the accusative Case assignment and the presence of *ha* in transitive verbal nouns (in Chung's terms, 2014:21). For this reason, Chung argues that *-(u)m* nominalizations of verbal nouns in Korean always target verbal structures at or above the *vP*.⁹⁾

- (29) a. ?John-uy tosi-uy/*lul phakoy-Ø
 John-GEN city-GEN/ACC destruction
 'John's destruction of the city'
 b. John-uy tosi-*uy/lul phakoy-ha-m
 John-GEN city-GEN/ACC destruction-*v*-NML
 'John's destruction of the city'

(Chung 2014:21)

Here we point out that the interpretation of the genitive argument is different in these two examples: in (29a) *John* is interpreted as the person to whom the destruction belongs. In (29b), *John* is interpreted as the possessor of the city. That is, the noun *phakoy* (destruction) is ambiguous between the event (the

9) Note that Chung gives conflicting judgements between our (29b) (his 31b) and his (25a). He judges the former to be fine and the latter to be ?. By contrast, the corresponding author of this paper finds this example ungrammatical, and we indicate the sentence as * in our own data later in the analysis.

former) and the result (the latter) reading (see Alexiadou 2001).¹⁰⁾ If *John* in (29b) is interpreted as an agent of the event of *destruction of the city*, it becomes ungrammatical despite the existence of *ha*.¹¹⁾

We realign Chung's data in (29) to remove the ambiguous reading, as in (30): the combination of GEN-*ki*, the direct object, and the verbal noun with light verb *ha*. In (30a), accusative Case assignment is impossible for the agent reading, as we discussed. The fact leads us to assume that there is no such correlation between *ha* and accusative case assignment. As such, the argument for or against the nominalizer taking vP as a complement disappears.

10) As cited in Alexiadou (2001), Picallo (1991) and others observe that a result reading and an event reading in Greek is distinguished in that the former can have the properties of being inconsistent/inexact, while the latter do not have such possibilities. (ia) is ungrammatical with predicates denoting the state of being inconsistent. This clearly shows the contrast on the result reading and the event reading in Greek.

- (i) a. *i paratirisi tu Jani epi dio ores itan anakrivis
 the observation the John-GEN for two hours was inexact
 'the observation of John for two hours was inexact'
 b. I paratirisi tu Jani itan anakrivis
 the observation the John-GEN was inexact
 'John's observation was inexact'
 c. I paratirisis tu Jani
 the observations the John-GEN
 'John's observations'

(Alexiadou 2001:37)

11) The grammaticality and the interpretation of the example in (29) is judged by nine native speakers. All answered as the phrase is interpreted as someone destroyed John's city. None of the speakers interpreted the genitive-marked noun as an agent.

- (30) a. John-uy tosi-(*lul) phakoy-ha-m
 John-GEN city-ACC destroy-v-NML
 ‘John’s destruction of the city’
 b. John-i tosi-(lul) phakoy-ha-m
 John-NOM city-ACC destroy-v-NML
 ‘that John destroys the city’

In (30), the accusative Case assignment is still not obligatory but optional when the external argument is marked with a nominative Case. Thus, there is no correlation between *ha* and the accusative Case assignment. Instead, we assume that *ha* is a dummy verb (Jo 2000), which does not license an accusative Case to its direct object when it follows the verbal noun. Eventually, we depart from Chung’s (2014) approach in assuming that the Case assignment on the direct object should be considered with the presence of Voice (Kratzer 1996).

To conclude, we have introduced the previous approaches relevant to Korean nominalization and problems with them. Given the various problems introduced above, we now turn to a full discussion on the syntactic structures of Korean *-ki* nominalization.

5. Proposal

This section offers an analysis of *-ki* nominalizations and compares them with better-known cases from English. In subsection 5.1, we concentrate on the projections of nominalizations which relies on the presence of VoiceP within their structure. In 5.2, we give a full structure of *-ki* nominalizations. In 5.3, we provide an account for the asymmetrical realizations of modification in nominalization.

5.1. Projections in *-ki* Nominalizations

Recall the three types of nominalizations in English under consideration, shown in (31). Kratzer, following a long line of earlier research (Abney 1987; Chomsky 1970; Johnson 1991, *inter alia*), proposes that gerundive nominalizations contain varying amounts of verbal structure. Kratzer argues that the presence of accusative Case in *ACC-ing* and *POSS-ing* indicates the presence of the VoiceP projection as Voice assigns accusative Case. The absence of accusative Case in *of-ing* constructions suggests that VoiceP is absent.

- | | |
|-------------------------------|-----------------|
| (31) a. Mary eating the apple | <i>ACC-ing</i> |
| b. Mary's eating the apple | <i>POSS-ing</i> |
| c. Mary's eating of the apple | <i>of-ing</i> |

In English, nominative subjects are not found in gerunds (in contrast to Korean), so VoiceP is taken to be the highest functional projection in English nominalizations.¹²⁾ Crucially, TP is not found in English nominalizations.

Recall that the two types of Korean *-ki* nominalizations differ with respect to the case marking on the subject as well as the object, as discussed in section 3.1. Therefore, we consider domains for the two different *-ki* nominalizations (NOM-*ki* vs. GEN-*ki*) below.

Following Kratzer's Voice theory, we can assume that the direct object in *-ki* nominalizations with NOM-*ki* (32) receive accusative

12) We set aside gerundive clausal supplements, which, in very formal English, have nominative subjects (Huddleston & Pullum, 2002). In casual speech the accusative form is used.

i) They appointed Max, [he/him being the only one who spoke Greek].

Case, and therefore, it must introduce an external argument. That is, the nominalized constituent must include VoiceP. In fact, we argue that *-ki* nominalizations with NOM-*ki* must project to at least TP as this type of nominalizations can have a past tense (32b), as we discussed in (8) and (10) in addition to nominative Case on the subject.¹³⁾

- (32) a. Mina-nun {Yunu-ka swukcey-*(lul) ceychwul-ha}-ki
 Mina-TOP Yunu-NOM homework-ACC submit-do-NML
 -man-ul kitaly-ess-ta.
 -DN-ACC wait-PAST-DEC
 'Mina waited for Yunu to submit his homework.'
 b. Mina-nun {Yunu-ka swukcey-*(lul)
 Mina-TOP Yunu-NOM homework-ACC
 ceychwul-ha-yess}-ki-ttaymwuney kipp-ess-ta.
 submit-do-PAST-NML-REASON happy-PAST-DEC
 'Mina was happy because Yunu (has) submitted his homework.'

On the contrary, the direct object in *-ki* nominalizations with GEN-*ki* does not receive accusative Case, as in (33). Therefore, the nominalized constituent does not contain VoiceP (Kratzer 1996).

- (33) {Yunu-uy chayk-(*ul) ilk}-ki-ka Mina-pota te nas-ta.
 Yunu-GEN book-ACC read-NML-NOM Mina-ADV better-DEC
 'Yunu's reading of the book is better than Mina's.'

The empirical differences between *-ki* nominalization with NOM-*ki* and GEN-*ki* are summarized in <Table 2>. The NOM-*ki*, as we have already discussed in (32), occurs at least TP including a VoiceP. The GEN-*ki*, the absence of the accusative case is accompanied by the absence of the verb's external argument. Consider now the interpretations of (33), the external argument of the predicate (*is better*) is not just *Yunu* but *Yunu's reading of the*

13) DN in (32a) is the abbreviation form of dependent noun in Korean.

book: the former can be the agent while the latter cannot. Thus, we propose that the domain for the GEN-*ki* must not include VoiceP where accusative Case is available.

<Table 2> Types of Nominalizations in Korean

				ACC	Voice
NOM- <i>ki</i>	[(Subject _{NOM} Object _{ACC} V]- <i>ki</i>			+	+
GEN- <i>ki</i>	[(Subject _{GEN} Object-Ø V]- <i>ki</i>			–	–

Before closing this section, we consider the above mentioned puzzling case of *-ki* nominalization in (6) repeated in (34). The example does not seem to have the correlation between nominative Case of external argument and accusative Case of the direct object, in that the accusative Case must be omitted although the external argument is introduced by nominative case assignment. The crucial problem here is that the external argument marked with the nominative case is expected to be realized at IP, while the direct object which cannot receive the accusative Case must be realized below VoiceP. However, it is not possible that a nominalized phrase is projected at two different projections simultaneously.

- (34) na-nun [Mina-ka pomwul-(*ul) chac-ki-(lul) ha]-ki-lul
 I-TOP Mina-NOM treasure search-NML-ACC do-NML-ACC
 pala-n-ta.
 hope-PRES-DEC
 'I want Mina to search for treasure.'

We assume that a possible way to account for the omission of accusative case in (34) is to say that the direct object *pomwul* and the nominalized verb *chac-ki* which lacks the function of accusative Case assignment combined as a noun complement of the

nominalized verb *ha-ki* (do-NML) (see also Chomsky 1956, 1970; Grimshaw 1990). The phenomenon by which a verb forms a compound with its direct object is known as noun incorporation (NI) (Mithun 1984). We argue that the direct object is incorporated before nominalization. Under NI, the nominalized NP in (34) can be roughly illustrated as in (35).

- (35) $[[_{NP} \text{Mina-NOM } [_{VP} [_{NP} [_{VP} t_i \text{ } [_{NP_i} V]] \text{ } [_N \text{-}ki]]\text{-ACC } [_V \text{do}]]]$
 $[_N \text{-}ki]]\text{-ACC}$

To conclude, we proposed internal projections for two types of Korean *-ki* nominalizations pinpointing Kratzer's Voice theory. In addition, we assume that the omission of the accusative Case in *-ki* nominalization with GEN-*ki* might be a possibility of NI. In the next section, we investigate the maximal projections of two types of nominalization and propose the full syntactic structure of nominalizations in Korean.

5.2. Proposal

Based on our observations and previous analyses, we argue here that Korean *-ki* nominalizations have two separate syntactic structures, one for NOM-*ki* and one for GEN-*ki*. Furthermore, we assume that *-ki* is an *n* head in Marantz's terms and is akin to English *-ing*.

First, we argue that *-ki* nominalizations with NOM-*ki* consist of an *n* that takes TP as a complement. Consider (36) repeated from (1a). The nominalized constituent must include VoiceP which introduces external argument due to the presence of the accusative case (Kratzer 1996) which is accompanied by the little *v*.

- (36) Yeungi-nun [Minswu-ka chayk-*(ul) ilk]-ki-lul wenha-n-ta.
 Yeungi-TOP Minswu-NOM book-ACC read-NML-ACC want-PRES-DEC
 'Yeungi wants Minswu to read books.'

Also consider (37) repeated from (32b), which shows that *-ki* nominalizations with NOM-*ki* can have a past tense. This supports our analysis that the nominalized constituent must include TP.

- (37) Mina-nun [Yunu-ka swukcey-*(lul) ceychwul-ha-ss]-ki
 Mina-TOP Yunu-NOM homework-ACC submit-do-PAST-NML
 -ttaymwuney kipp-ess-ta.
 -REASON happy-PAST-DEC

In order to explicitly state our analysis of *-ki* nominalization with NOM-*ki*, we propose the hierarchy of projections as in (38). Note that we assume that full nominal phrases are KP (given overt case marking), although it does not bear on our analysis. The schema for the nominalized constituent in (36) is also described as in (39).

- (38) Hierarchy of Projections for *-ki* nominalization with NOM-*ki*
 $nP < TP < \text{VoiceP} < vP < \sqrt{P} < KP$

- (39) Schema for (36)

$[_{nP} [_{n'} [_{TP} [_{KP} \text{Minswu-NOM}]_i [_{T'} [_{T}] [_{\text{VoiceP}} t_i [_{\text{Voice}}] [_{vP} [_{v'} [_{v}]$
 $[_{\sqrt{P}} [_{KP} \text{book-ACC}] [_{\sqrt{v}} \text{read}}]]]]]]]] [_{n} \text{-ki}]]]$

In (39), the KP complement of the nominalized verb is marked with an accusative case by Voice, and the past tense is allowed by virtue of the presence of the TP. In addition, the presence of Voice explains the external argument is the AGENT of the nominalized verb.

Contrastively, we argue that *-ki* nominalizations with GEN-*ki* contain *nP* followed by *vP* and $\sqrt{P}$ respectively, and the *nP* follows

DP for the genitive Case assignment to the external argument. Consider (40) repeated from (1b). As we discussed, the absence of an accusative case accompanies the absence of the VoiceP where an accusative Case is available.

- (40) Minswu-uy chayk ilk]-ki-ka sicak-toy-ess-ta.
 Minswu-GEN book read-NML-NOM start-PASS-PAST-DEC
 'Minswu's reading a book has started.'

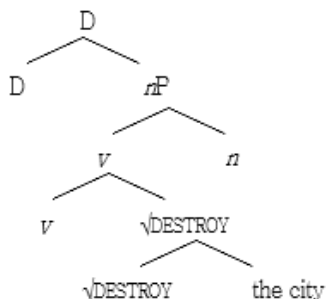
One outstanding issue is the theta role of the external argument. Kratzer argues that the external argument of *of-ing* constructions in English is free. (*cf. Maria's reading of Pride and Prejudice*). Maria is not necessarily the agent of *read* in this example. In Korean, the theta-relation is not free, however, a fact which is not predicted by the analysis here. We do not present a complete solution to this problem, but suggest that Moulton (2004) might provide a solution. Moulton points out certain situations, even in *of-ing* constructions in English where the external argument is obligatorily interpreted as the agent of the event. He proposes that such constructions have a "defective *v*" (a defected Voice in our terms). This defective head introduces an external argument, but does not assign accusative Case. Although the forthcoming analysis does not pursue this idea in detail, we adopt the spirit of this approach, pending further research (Chung, forthcoming).

Consider the following example in (41a) in which the past tense is not allowed unlike *-ki* nominalization with NOM-*ki* (41b). This fact supports our analysis that *nP* for *-ki* nominalization with GEN-*ki* does not include TP.

- (41) a. Mina-nun [Yunu-uy chayk ilk-(**ess*)]-ki-ttaymwun-ey
 Mina-TOP Yunu-GEN book read-PAST-NML-REASON
 kippu-ta.
 be.pleased-DEC
- b. Mina-nun [Yunu-ka chayk-ul ilk-ess]-ki-ttaymwun-ey
 Mina-TOP Yunu-NOM book-ACC read-PAST-NML-REASON
 kippu-ta.
 be.pleased-DEC
 'Mina is pleased with Yunu to have read a book.'

To determine the projection for the nominalized verb, we adopt Marantz's (2001) analysis that nominalizations are not derived from a lexicon (*contra* Chomsky's *Remarks on Nominalization*), rather, roots are underspecified for lexical categories. In (42), consider how a noun is created from a verb above little *v* under Marantz's analysis.

- (42) John's destroying the city/John's destroying of the city



(Marantz 2001:26)

Following Marantz (2001), we propose that the nominalized verb *ilk-ki* (read-NML) in (40) is created from a verb above little *v*: the nominalizer *-ki*, as an *n*, is merged with the root $\sqrt{ilk}$ ($\sqrt{\text{read}}$) before merging another functional head. Then, the hierarchy of projections for *-ki* nominalizations with GEN-*ki* is created as in

(43) including the $\sqrt{P}$. Note that we assume that the object is a bare DP rather than a KP given the obligatory absence of case marking. The schema for the nominalized constituent in (40) is described as in (44).

- (43) Hierarchy of Projections for *-ki* nominalization with GEN-*ki*
 DP $\succ$ nP $\succ$ vP $\succ$ $\sqrt{P}$ $\succ$ DP

- (44) Schema for (40)
 $[_{DP} [_{DP} \text{Minswu-GEN}]_i [_{D'} [_{nP} [t_i [_{vP} [_{\sqrt{P}} [_{DP} \text{book}] [_{\sqrt{V}} \text{read}]]] [_{v}]]] [_n \text{-ki}]] [_D]]]]]$

In (44), the accusative Case marking on the DP complement of the root verb is impossible due to the absence of Voice. The past tense marking is also impossible due to the absence of the TP.

In conclusion, the structure of Korean *-ki* nominalizations can be distinguished from the existence of VoiceP and TP. This is determined by the Case marking on the external argument of *-ki* nominalization. Specifically, we argue that NOM-*ki* takes TP as a complement, while GEN-*ki* takes vP as a complement.

5.3. Modifiers

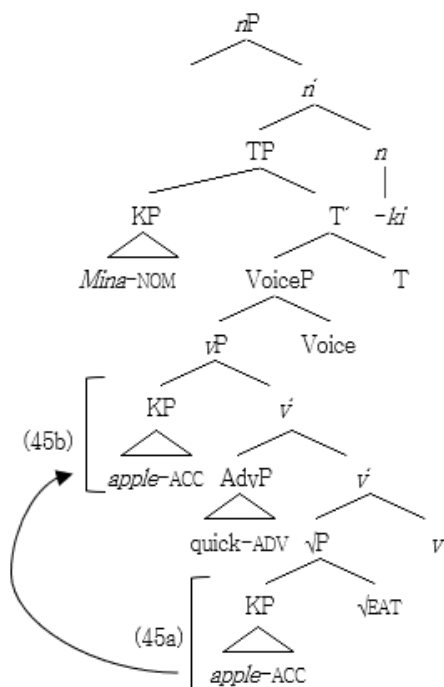
In this section, we provide an account of the asymmetric modification system under NI. Consider first the following examples in (45), repeated and simplified from (11), modification pattern in NOM-*ki*.

- (45) a. Mina-ka *ppalu-n/ppalu-key sakwa-lul mek-*ki*
 Mina-NOM quick-ADN/ADV apple-ACC eat-NML
 ‘that Mina quickly eats an apple’

- b. Mina-ka sakwa-lul *ppalu-n/ppalu-key mek-ki
 Mina-NOM apple-ACC quick-ADN/ADV eat-NML
 'that Mina quickly eats an apple'

We argue that adnominals are not found inside nominalizations with NOM-*ki* as there are only verbal projections, where only adverbs are allowed. Also, we claim that the direct object optionally scrambles out of the $\sqrt{P}$ to Spec of vP to account for the word order difference in between (45a) and (45b). The structure for (45) is illustrated in (46).

- (46) The position for adnominal/adverb in NOM-*ki* nominalization



When the direct object *sakwa* (apple) does not move out of $\sqrt{P}$, we get the nominalized phrase like (45a), as presented in (46) where the adverb precedes the direct object. When the direct object moves from $\sqrt{P}$ to Spec of vP , the direct object is followed by the adverb.

Now consider modification patterns in *-ki* nominalizations with GEN-*ki* in (47), repeated and simplified from (12). Adverbs in both are grammatical, while the only adnominal not intervening but preceding direct object is grammatical.

- (47) a. Yunu-uy ppalu-n/ppalu-key sakwa mek-*ki*
 Yunu-GEN quick-ADN/ADV apple eat-NML
 ‘Yunu’s quick/quickly eating an apple’
 b. Yunu-uy sakwa *ppalu-n/ppalu-key mek-*ki*
 Yunu-GEN apple quick-ADN/ADV eat-NML
 ‘Yunu’s quick/quickly eating an apple’

Here we address why the preceding the direct object adnominal is grammatical (47a) and the intervening one is ungrammatical (47b). We propose the adnominal in *-ki* nominalizations with GEN-*ki* is as in (48a), which gives a rough syntactic structure of (47a). We can observe two possible nominal categories that the adnominal could target: either the direct object *sakwa* (apple) or the nominalized phrase *sakwa mek-ki* (eating an apple). Excluding the former *ppalu-n sakwa* (quick-ADN apple) which is semantically ridiculous, the only possible nominal category to be modified by adnominal is the latter. In contrast, the direct object and the nominalized verb in (45a) do not form an NP but a VP where adnominal modification is impossible as in (48b).

- (48) a. [Yunu-uy [ppalu-n [_{nP} [_{vP} sakwa mek]-ki]]]
 Yunu-GEN quick-ADN apple eat-NML
 'Yunu's quick eating an apple'
- b. *([_{nP} [Mina-ka [_{VOICEP} [_{vP} [ppalu-n [_{vP} sakwa-lul mek]]]])]
 Mina-NOM quick-AND apple-ACC eat
 [_n -ki])
 -NML
 'that Mina quickly eats an apple'

Assuming a more articulated structure, we suggest the possibility of noun incorporation (NI) (see Gerdts 1998; Massam 2009; Barrie & Mathieu 2016) in *-ki* nominalizations with GEN-*ki* based on two things as follows. First, the accusative Case of the direct object is omitted. Second, adnominal modifiers cannot appear lower than the direct object.

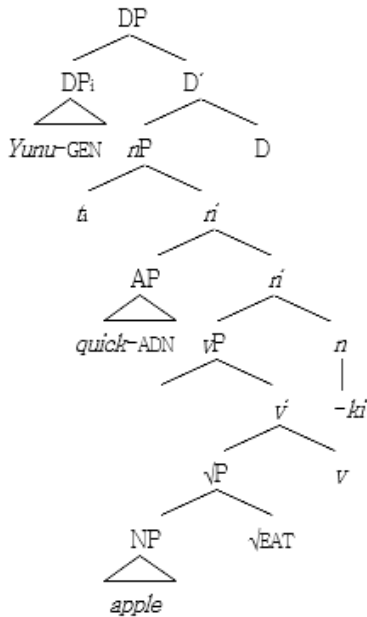
We propose the structure for (47a) as in (49), which shows that NI occurs. In (49), the only projection which allows adnominal is *nP*, and it is higher than the position of the direct object.¹⁴⁾ Therefore, there is only one possibility of the surface position for adnominal in *-ki* nominalization with GEN-*ki*: above VP. In addition, the adnominal cannot appear between the direct object and the verb, which is already in the form of NI. As for adverbial modifications, we can simply rule out with the fact that there is no possible projection for them inside the *nP*.

The astute reader will notice that there is no theta-position for the argument, *Yunu*. Under Kratzer's original proposal, this argument does not receive an obligatory agent theta-role; however, as we mentioned above only an agent reading seems to be available

14) One question we need to address is why the *nP* in (46) cannot be the position for adnominals, while it must be in (49). Note that the external argument must precede all other elements inside the nominalization. Therefore, if the adnominal occurs at *nP* above TP in (46), then it cannot get the right word order, preceding the external argument as well as other elements.

in Korean. We suggested above that Moulton (2004) might provide a solution for us if we assume a defective VoiceP that licenses an external argument but does not assign accusative Case. Again, we leave the implementation of this idea to future research.

(49) The position for adnominal in GEN-*ki* nominalization



In this section, we suggest an explanation for the modification asymmetry accompanied by the difference of the Case on the external argument. We proposed the movement of the direct object and the occurrence of NI inside a nominalization in order to analyze the asymmetrical patterns on modification in nominalizations.

6. Conclusion

We have examined the syntax of *-ki* nominalizations in Korean, noting several similarities with English gerundive constructions. In particular, we have noted that there are two kinds of *-ki* nominalizations in Korean: NOM-*ki* and GEN-*ki*. In the NOM-*ki* construction, the subject has nominative Case, the object has accusative Case, both present and past tense marking is available, and only adverbial modifiers are permitted. In the GEN-*ki* construction, the subject has genitive Case, the object is caseless, past tense marking is not available, and both adverbial and adnominal modifiers are found, but are constrained by linear order. We proposed an analysis in the vein of Kratzer (1996), updated to take into account Marantz's proposal on categorizers (*n* and *v*) and the notion that the vP layer is split into two projections: VoiceP and vP. The NOM-*ki* construction consists of the nominalizer *-ki* taking a TP as a complement, while the GEN-*ki* construction consists of *-ki* taking a vP as a complement. These two structures account for the constellation of properties discussed in this paper.

In addition to adding to the general cross-linguistic investigation of the nominalization of clausal and verbal constituents, this study has shown that clausal nominalizations can vary more than previously thought. Alexiadou (2001) proposes that AspP is the highest projection in the extended verbal projection that can be nominalized. The Korean facts show quite clearly that TP can be nominalized as both nominative subjects and past tense marking are found in the NOM-*ki* constructions. We leave the cross-linguistic implications of this conclusion to future research.

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Barrie Michael

Department of English

Sogang University

Associate Professor

35, Baekbeom-ro, Mapo-gu, Seoul, Republic of Korea [04107]

E-mail: mikebarrie@sogang.ac.kr

Heeryun Chung

Department of English

Sogang University

PhD Student

35, Baekbeom-ro, Mapo-gu, Seoul, Republic of Korea [04107]

E-mail: cheeryun2@gmail.com

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